ServerCats

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Agenda

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5. Tools, Technologies and Linux Environment Used
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AUTOMATED LINUX-BASED COLLEGE LABORATORY MANAGEMENT SYSTEM

Problem Statement

Manual management of laboratory users, departmental storage, system monitoring, and backups is time-consuming, inefficient, and prone to errors.

Objective

What Server Cats. does

Server Cats is a Linux-based laboratory management system that automates server administration and monitoring.

- Creates and manages student accounts.
- Organizes departmental groups and shared storage.
- Monitors CPU, RAM, and disk utilization.
- Automates temporary file cleanup and backups.
- Displays system information through a web dashboard.

Solution

Server Cats provides an automated Linux-based solution for efficient laboratory administration and storage management.

- Automates student account creation and group management.
- Provides secure departmental storage with controlled permissions.
- Monitors system resources and disk utilization.
- Performs automated cleanup, backup, and recovery.
- Integrates a web dashboard for centralized system monitoring.

Project Overview

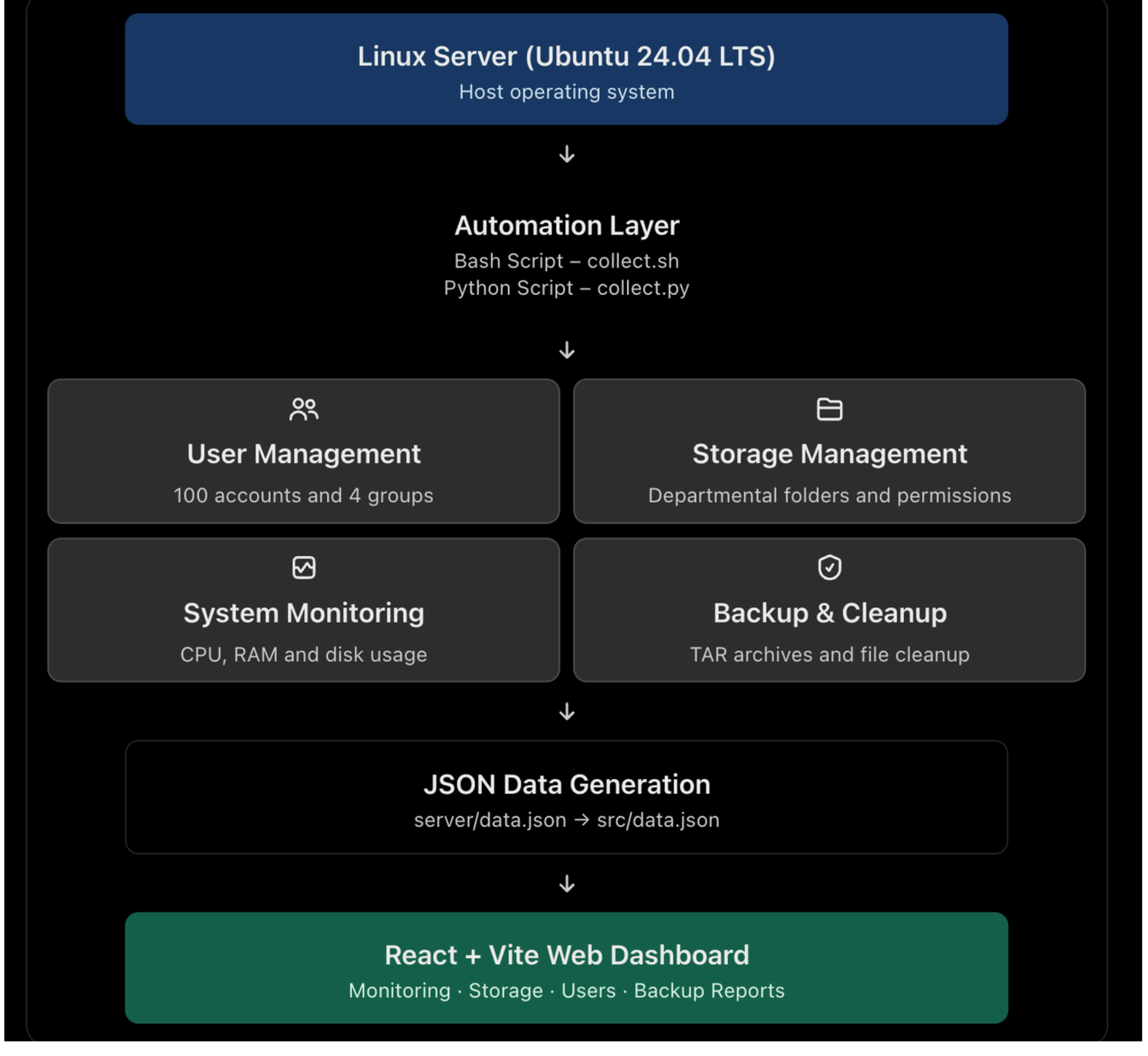
Server cats is a Linux-based laboratory management system designed to simplify and automate server administration.

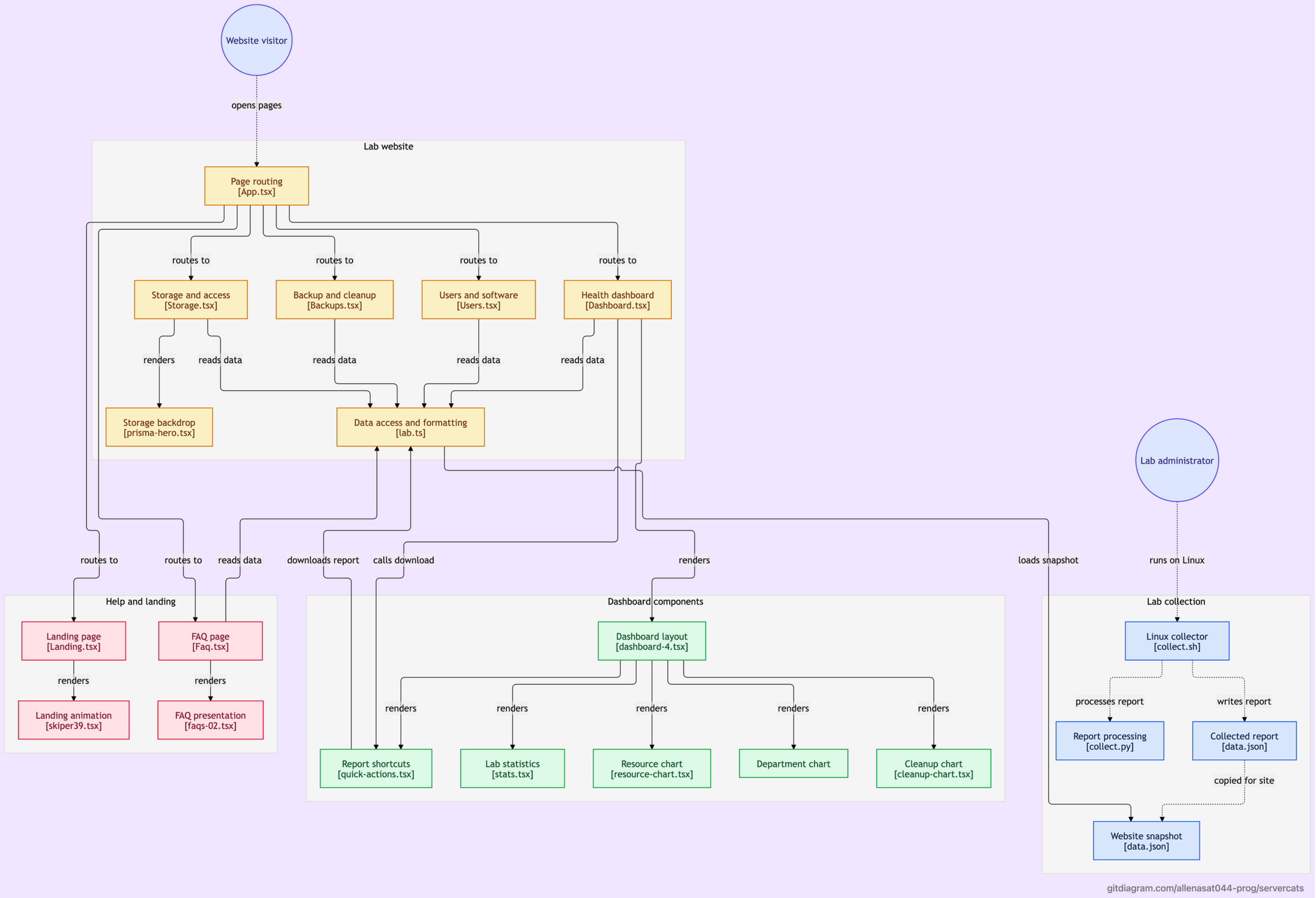
- Manages student accounts and departmental groups.
- Organizes shared storage with access permissions.
- Monitors server resources and system health.
- Automates backups, recovery, and file cleanup.
- Provides a React-based dashboard for centralized monitoring.

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SYSTEM ARCHITECTURE

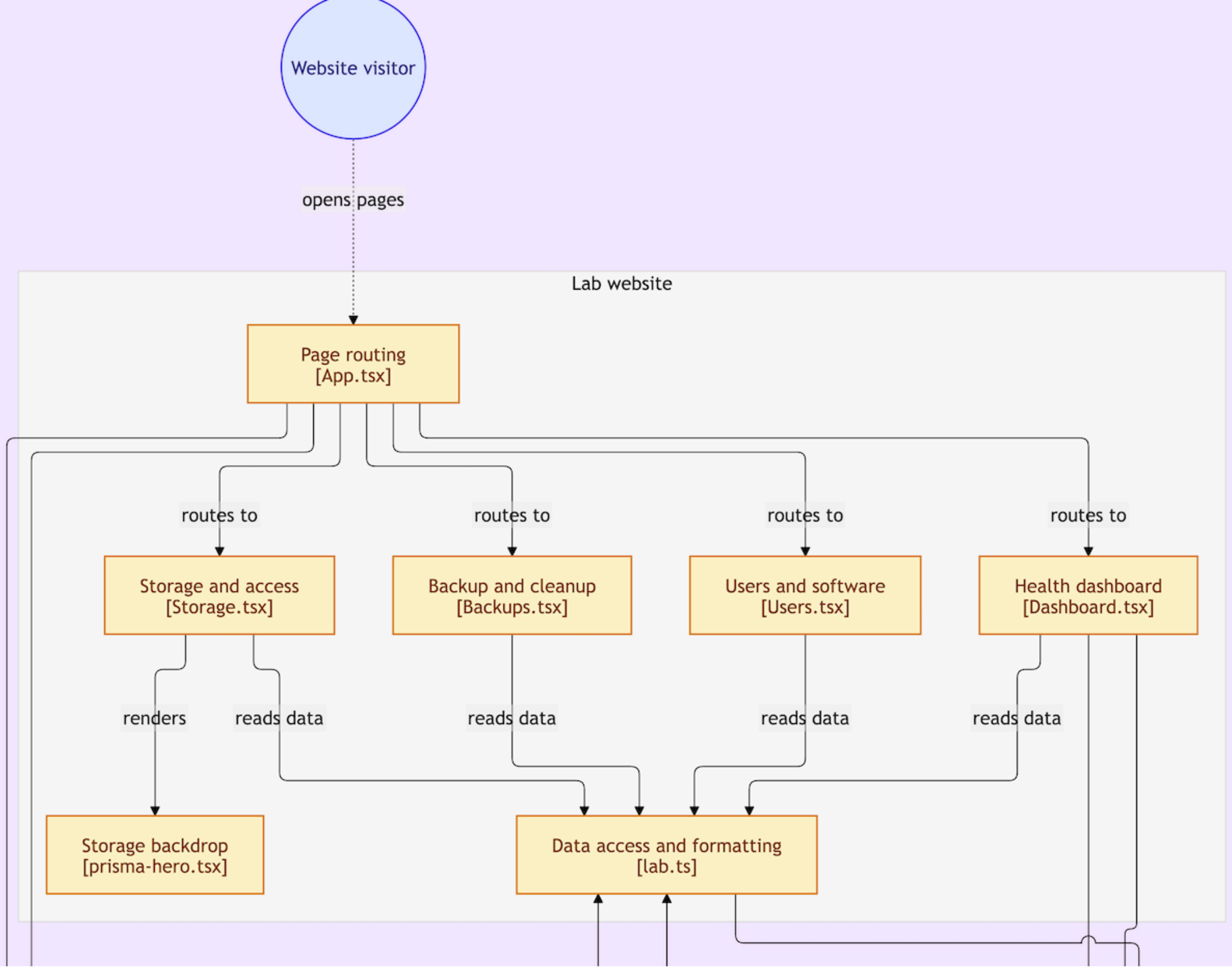


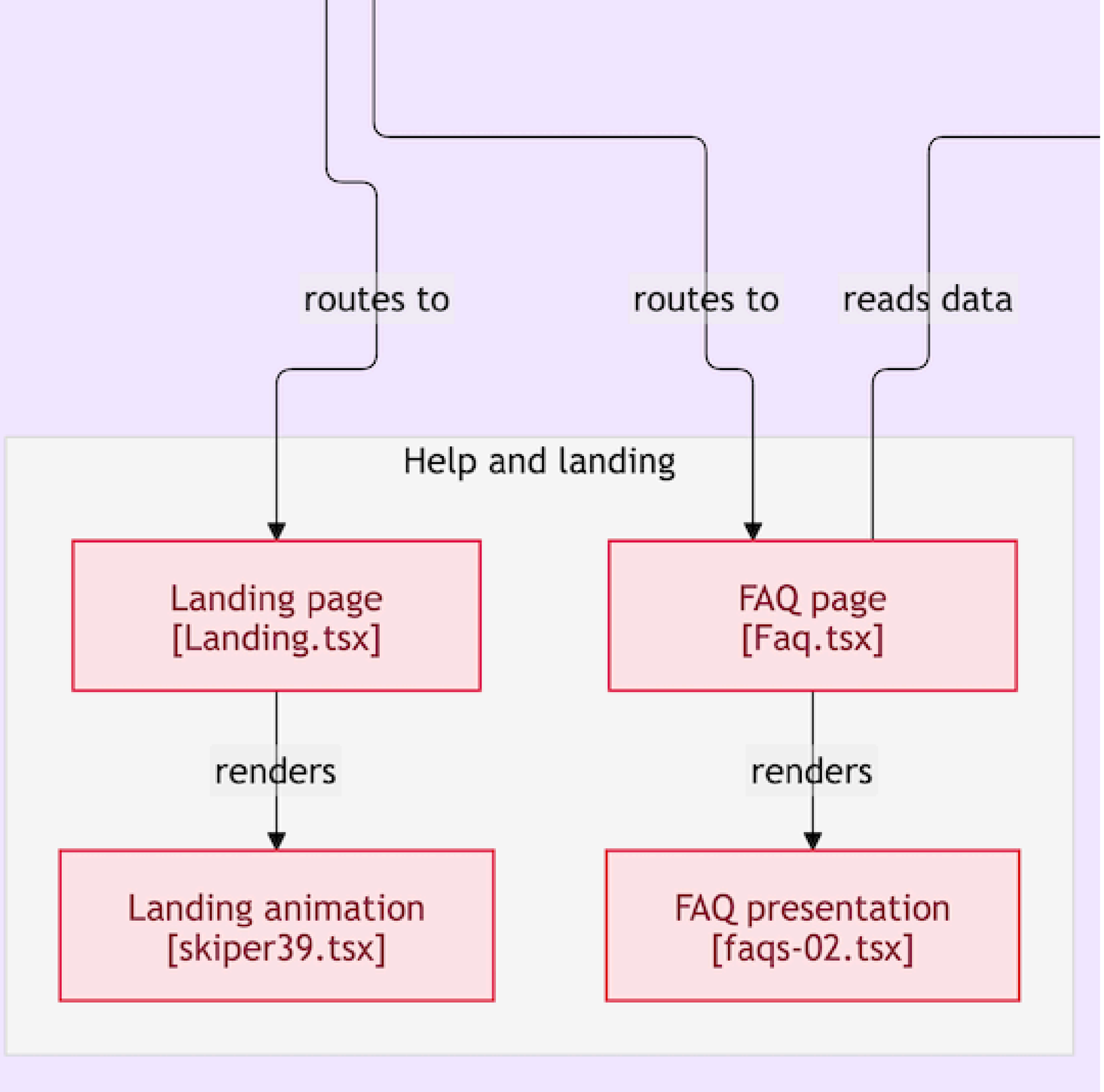

Linux Server (Ubuntu 24.04 LTS)
Host operating system
Automation Layer
Bash Script – collect.sh
Python Script – collect.py
User Management
100 accounts and 4 groups
Storage Management
Departmental folders and permissions
System Monitoring
CPU, RAM and disk usage
Backup & Cleanup
TAR archives and file cleanup
JSON Data Generation
server/data.json → src/data.json
React + Vite Web Dashboard
Monitoring · Storage · Users · Backup Reports



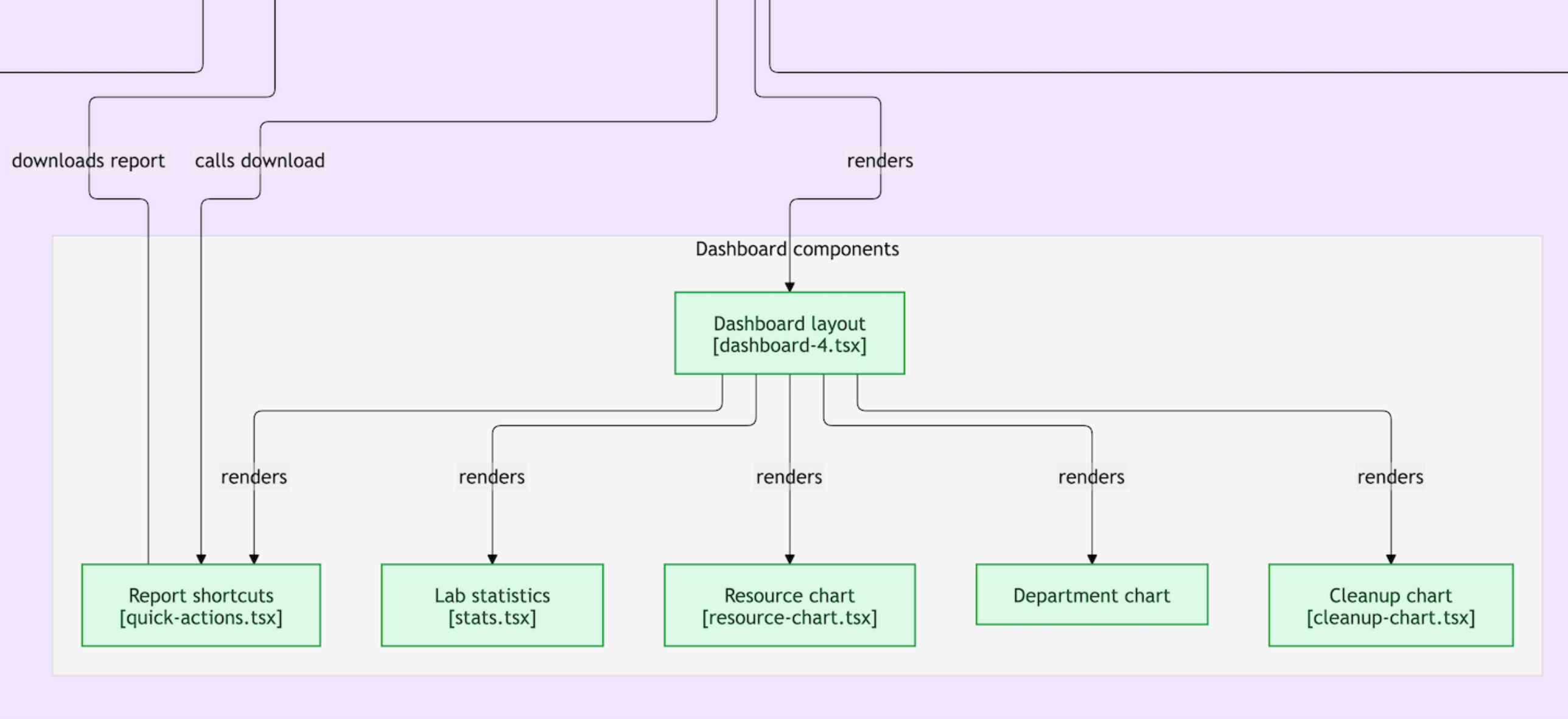
gitdiagram.com/allenasat044-prog/servercats

Lab Website

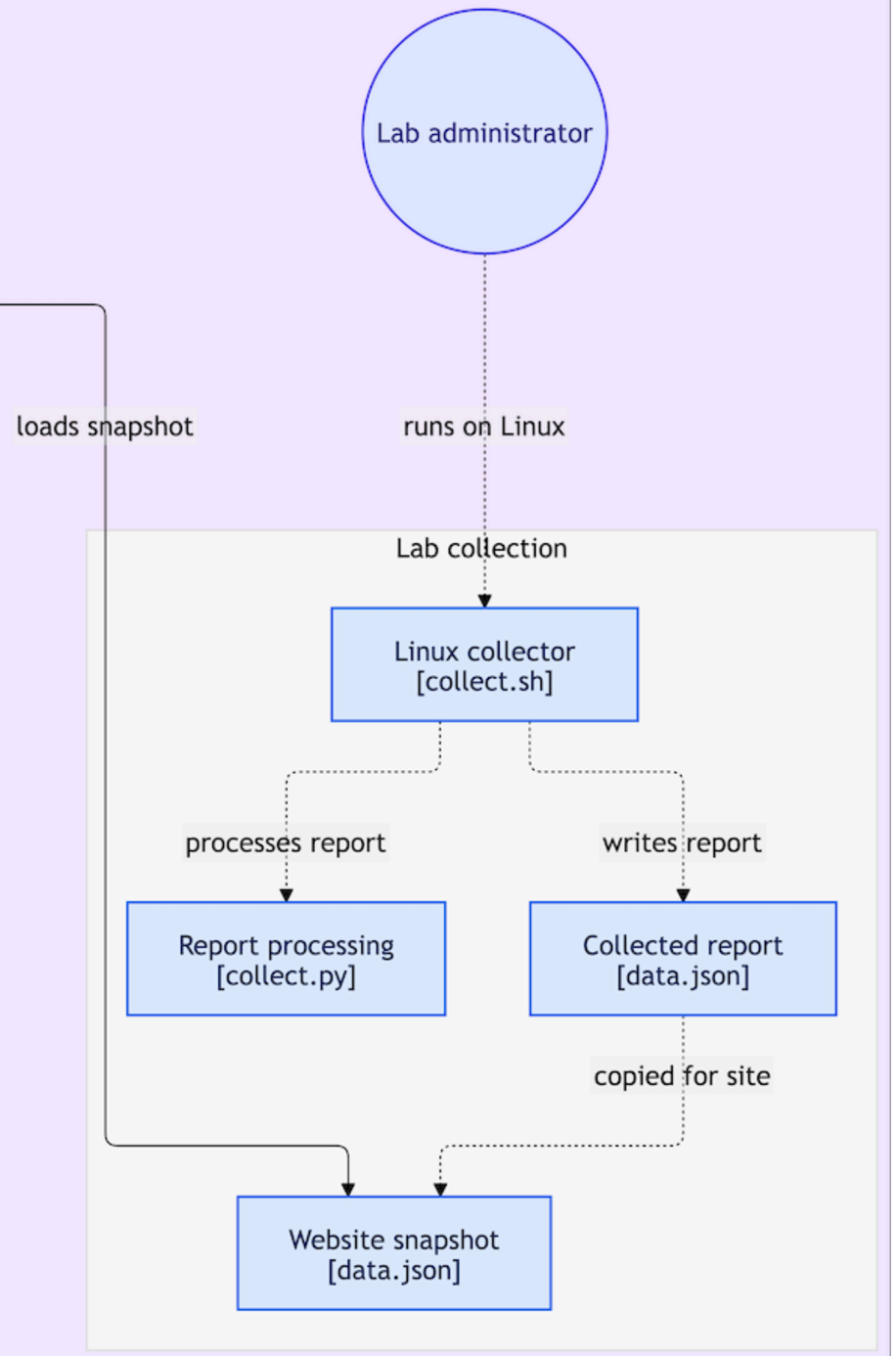

Website visitor
opens pages
Lab website
Page routing
[App.tsx]
routes to
routes to
routes to
routes to
Storage and access
[Storage.tsx]
Backup and cleanup
[Backups.tsx]
Users and software
[Users.tsx]
Health dashboard
[Dashboard.tsx]
renders
reads data
reads data
reads data
reads data
Storage backdrop
[prisma-hero.tsx]
Data access and formatting
[lab.ts]



Help &Landing



Dashboard Components



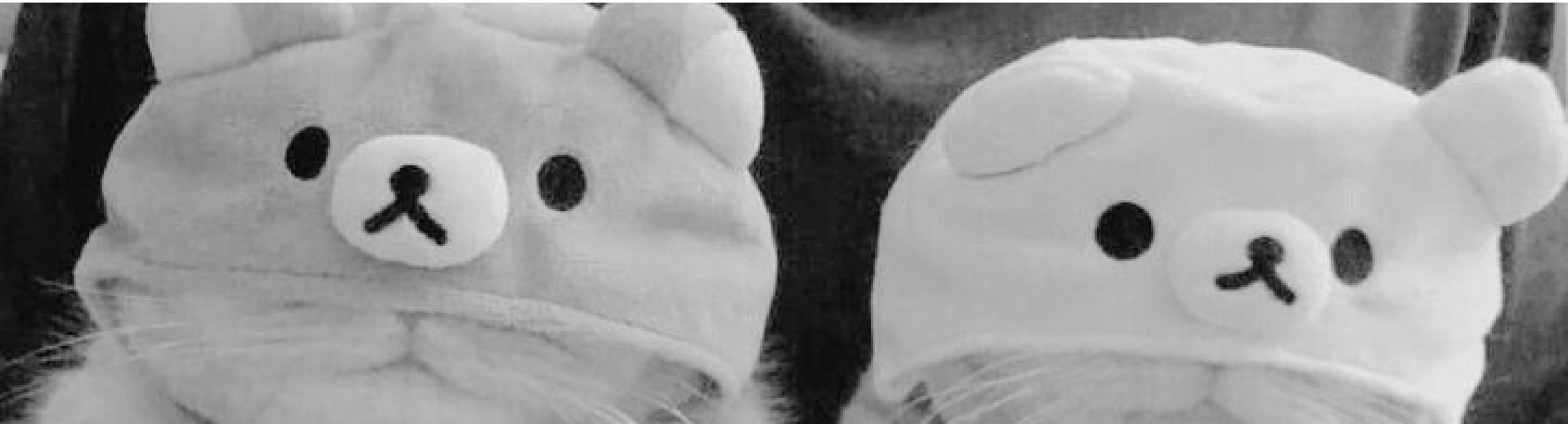
Lab Collections

Tools, Technologies and Linux Environment Used

- Operating System: Ubuntu 24.04 LTS
- Scripting: Bash Shell, Python 3
- Frontend: React.js, TypeScript
- Build Tool: Vite
- Styling: Tailwind CSS
- Data Exchange: JSON
- Backup Tools: TAR, GZIP, SHA-256
- Version Control: Git and GitHub
- Linux Utilities: useradd, groupadd, chmod, chown, df, free, find, ps

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LINUX COMMANDS & SCRIPTS



LINUX COMMANDS & SCRIPTS

The script writes students.csv (header username,department, then student001,cse ... student100,civil). Departments rotate CSE, ECE, MECH, CIVIL, so each has 25 students.

```
groupadd lab-cse
useradd -m -s /bin/bash -g lab-cse student001
echo "student001:Lab@2026" | chpasswd
chage -d 0 student001
```

```
getent passwd | grep -c '^student'
id student001
getent group lab-cse
chage -l student001
```

7.2 Department folders and permissions

```
chown :lab-cse /srv/lab/srv/cse
chmod 2770 /srv/lab/srv/cse
stat -c %a /srv/lab/srv/cse
```

TEMPORAL FILE CLEANUP & BACKUP /RESTORE TEST

7.4 Temporary-file cleanup

The script creates 30 files of 20 KB in `/srv/lab/tmp` (20 dated 10 days ago, 10 dated one day ago) and then removes everything older than seven days:

```
head -c 20480 /dev/urandom > /srv/lab/tmp/example.tmp
touch -d "10 days ago" /srv/lab/tmp/example.tmp
find /srv/lab/tmp -type f -mtime +7 -delete
```

7.5 Backup and restore test

```
tar -czf /srv/lab/backups/backup-20261015-031647.tar.gz -C /srv/lab srv
sha256sum /srv/lab/backups/backup-20261015-031647.tar.gz
mkdir -p /tmp/restore
tar -xzf /srv/lab/backups/backup-20261015-031647.tar.gz -C /tmp/restore
find /tmp/restore/srv -type f | wc -l
```

Storage Management / LVM / Backup and Recovery

- **8.2 Backup strategy**

Each run archives /srv/lab/srv into a time-stamped tar.gz, records its size, the first 16 characters of its SHA-256 checksum and the number of files, and keeps the six newest archives in the history shown on the website.

- **8.3 Restore test**

After creating the archive the script extracts it into a temporary folder, counts the files and compares the count with the original. Equal counts give PASSED, otherwise FAILED. The temporary folder is removed afterwards.

8.4 Recovery procedure

List the backups, restore into a safe folder, compare with the original, then copy back what is needed:

```
ls -lh /srv/lab/backups
mkdir -p /tmp/restore
sudo tar -xzf /srv/lab/backups/backup-20261015-031647.tar.gz -C /tmp/restore
diff -r /srv/lab/srv /tmp/restore/srv && echo "no differences"
```

LVM

```
sudo pvcreate /dev/sdb
sudo vgcreate vg_lab /dev/sdb
sudo lvcreate -L 50G -n lv_lab vg_lab
sudo mkfs.ext4 /dev/vg_lab/lv_lab
sudo mkdir -p /srv/lab
sudo mount /dev/vg_lab/lv_lab /srv/lab
```

Automation and Shell Scripting

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9.2 How the Bash script is built

Technique	Why
`set -u`	Stops on use of an undefined variable
`cd "$(dirname "$0")"`	Works from any folder
Root test `[ "$(id -u)" -eq 0 ]`	Chooses `/srv/lab` (root) or `$HOME/lab-server` (normal user)
Arrays `DEPTS`, `TOOLS`, `ROWS`	Loops over departments, tools and students
Guards (`getent group`, `id`)	Makes every run repeatable without errors
Helper `j()`	Escapes text before it is written into JSON
Helpers `gb()`, `hs()`	Format bytes as GB, MB or KB
Function `cpu_pct()`	Reads `/proc/stat` twice to compute CPU usage
Here-document `cat > data.json <<EOF`	Writes the JSON file

Security and User/Permission Management

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Control	Detail
Group-based access	Each department folder is mode `2770` and owned by its own group; other students have no access
Forced password change	`chage -d 0` expires the initial password so students must choose their own at first login
Separate roles	Only root creates accounts; a normal user run creates none
Safe scripting	`set -u`, quoted variables and an escaping function for JSON output
Read-only website	The dashboard only displays saved data; it has no function that changes the server
Tested backups	Every backup is restored to a temporary folder and verified

Check access control:

```
ls -ld /srv/lab/srv/*
id student001
getent group lab-cse
```

Conclusion and Future Enhancements

Project Summary

- Automates 100 student accounts and department groups.
- Manages folder permissions, software checks and temporary file cleanup.
- Performs verified backups and monitors server health.
- Displays results through a React dashboard.

Future Enhancements

- LDAP / Active Directory integration.
- Student-wise disk quotas.
- College database integration.
- Remote backups and automated alerts.
- Live dashboard API.
- Multi-lab support.

Home | Dashboard | Backup & Maintenance | Storage & Access | Users & Software | FAQ

Dashboard

Ubuntu 24.04.4 LTS · up 0 minutes

CPU usage
1%
1 core(s) · load 0

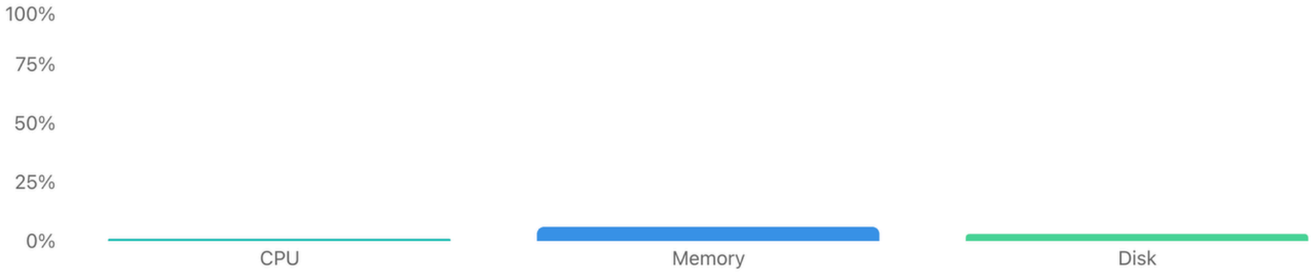
Memory
6%
236.0 MB of 3.9 GB

Disk space (/)
3%
8.5 GB of 252.0 GB

Logged in now
0
2 accounts running processes

Resource usage

Collected 15 Oct 2026, 03:16:47 on vm



Last cleanup

20 of 30 temp files removed · 400 KB freed



Students by department

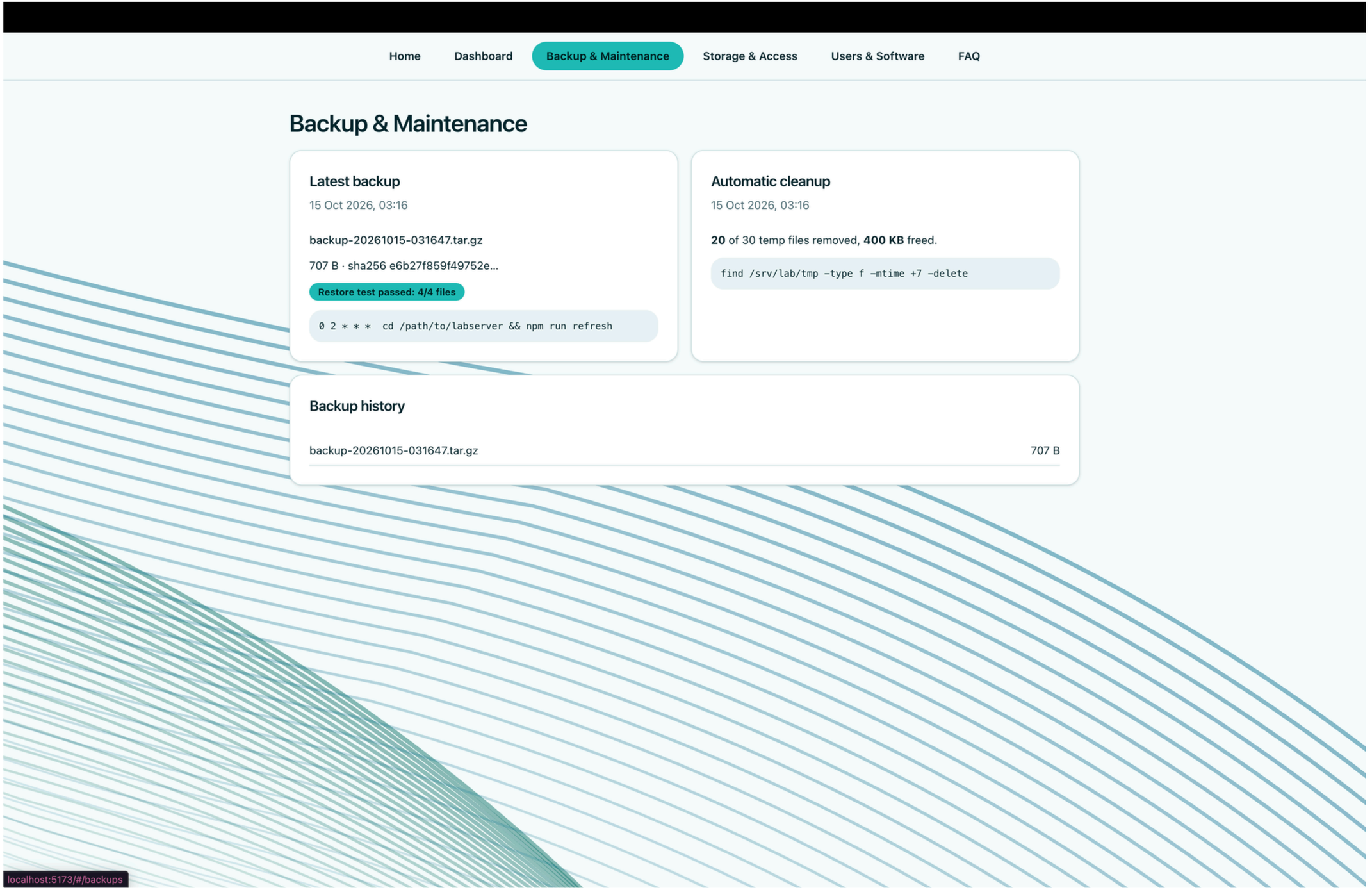
100 accounts

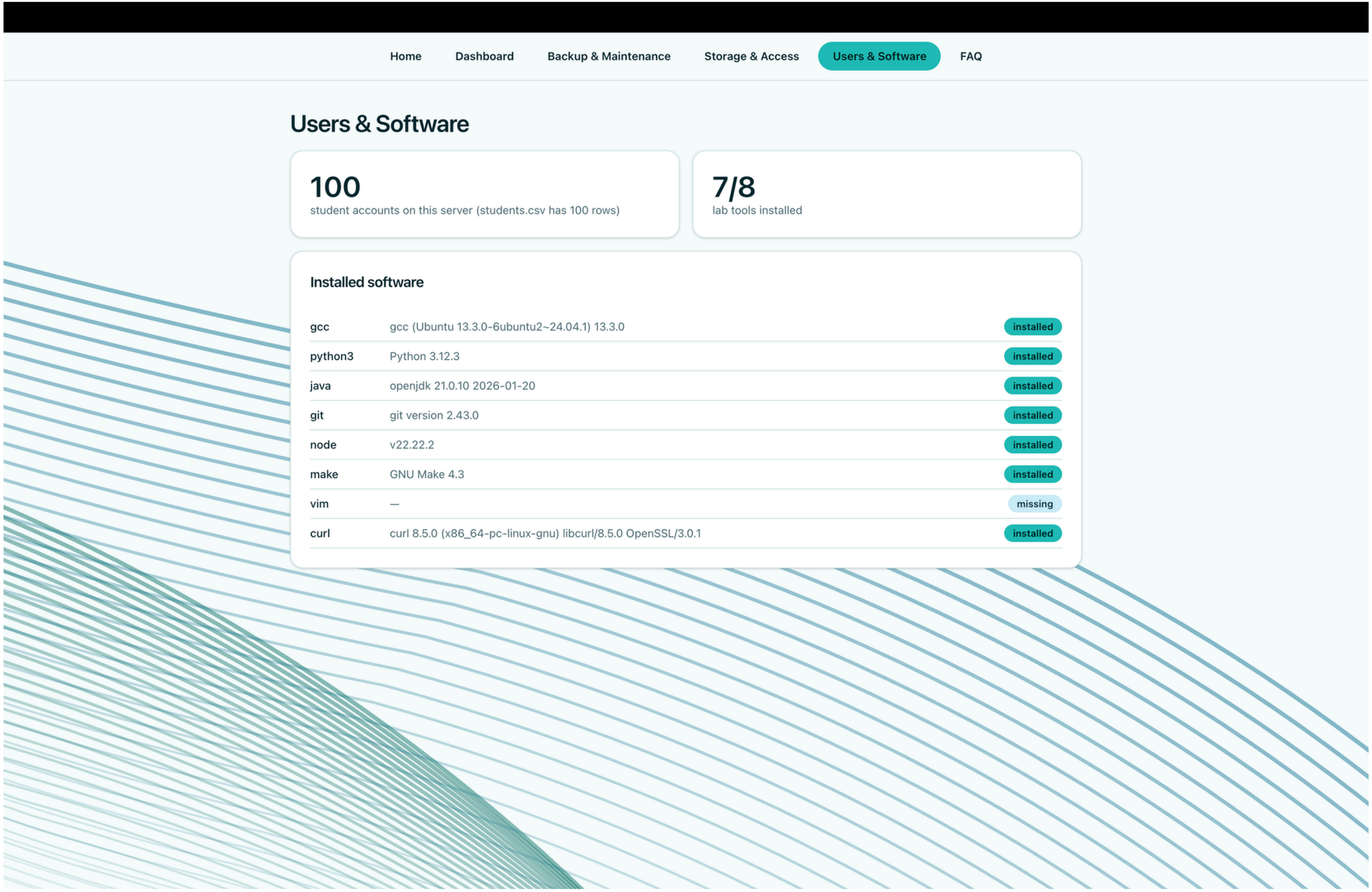
CSE	25
ECE	25
MECH	25
CIVIL	25

Quick actions

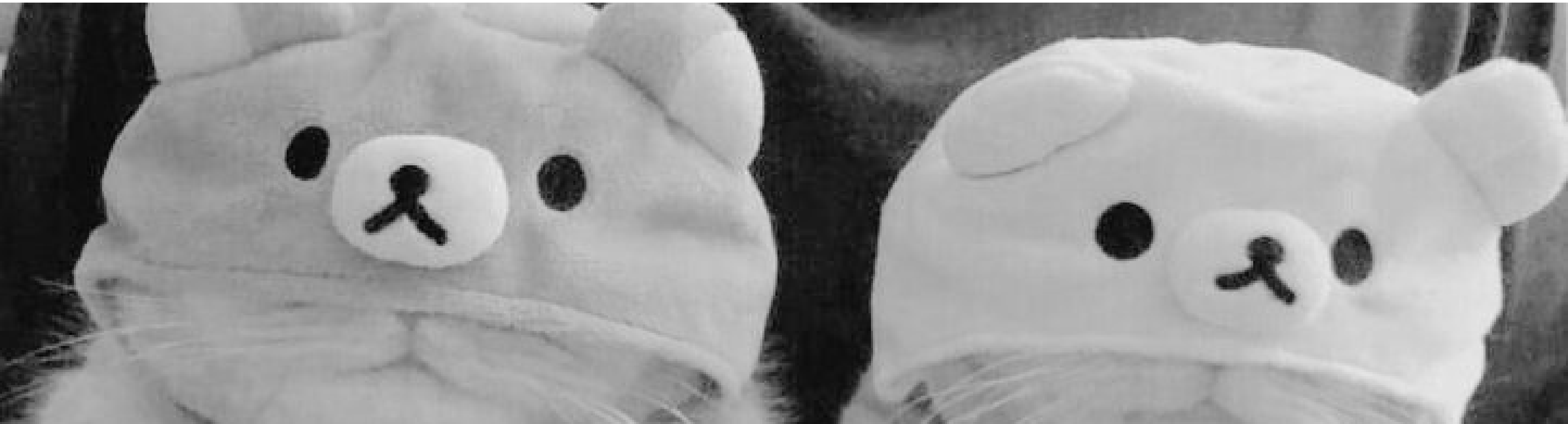
Report and shortcuts.

- Download report
- Backups & cleanup
- How to use


Home
Dashboard
Backup & Maintenance
Storage & Access
Users & Software
FAQ
Backup & Maintenance
Latest backup
15 Oct 2026, 03:16
backup-20261015-031647.tar.gz
707 B · sha256 e6b27f859f49752e...
Restore test passed: 4/4 files
0 2 * * * cd /path/to/labserver && npm run refresh
Automatic cleanup
15 Oct 2026, 03:16
20 of 30 temp files removed, 400 KB freed.
find /srv/lab/tmp -type f -mtime +7 -delete
Backup history
backup-20261015-031647.tar.gz
707 B
localhost:5173/#/backups


Home
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Backup & Maintenance
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Users & Software
FAQ
Users & Software
100
student accounts on this server (students.csv has 100 rows)
7/8
lab tools installed
Installed software
gcc
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
installed
python3
Python 3.12.3
installed
java
openjdk 21.0.10 2026-01-20
installed
git
git version 2.43.0
installed
node
v22.22.2
installed
make
GNU Make 4.3
installed
vim
—
missing
curl
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.1
installed

TESTING AND VALIDATION



Challenges Faced and Solutions

Challenge	Solution in the project
The script needs Linux (`/proc` , GNU `stat`) but the team develops on macOS and Windows	The website runs everywhere; the script runs in a Linux VM or WSL, and `data.json` is copied back with `scp`
Running as root creates real users, which is risky on a personal machine	The script detects root; as a normal user it uses `$HOME/lab-server` and creates no accounts
Creating 100 users must not fail on a second run	`getent group` and `id` guards skip what already exists
Cleanup needs old files to demonstrate	The script creates test files and ages them with `touch -d "10 days ago"`
Writing valid JSON from Bash is error-prone	A helper function `j()` escapes backslashes and quotes before they are written
The website has no live backend	The script saves `data.json` ; `npm run refresh` copies it into the website
A backup is not trustworthy until restored	Every run extracts the archive and compares file counts
Shared department files must stay accessible to the group	Mode `2770` with setgid so new files inherit the group
Different commands on different systems to view the saved report	Section 7.8 lists the Python, `plutil` (macOS) and PowerShell versions

THANK YOU